Arthur **Fournier**

Machine Learning · Robotics · Perception

Paris, France · Seeking an internship from April 2027

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Education

Mines Paris – PSL — *Paris, France*

Double degree with École polytechnique · Robotics track — *Sep 2026 – Dec 2027*

- Nonlinear control of manipulators and mobile robots, reinforcement learning for robotics, estimation and filtering, 3D vision.

Master MVA – ENS Paris-Saclay — *Paris, France*

Auditing student (non-degree) · 3D computer vision, robotics, point clouds and 3D modeling — *Sep 2026 – present*

École polytechnique — *Palaiseau, France*

MSc in Engineering (Ingénieur Polytechnicien program) — *2023 – 2026*

- Machine learning, deep learning, reinforcement learning, generative AI, computer vision, statistics, optimization and control.

Lycée Hoche — *Versailles, France*

Preparatory classes in Mathematics, Physics and Computer Science (MPI) — *2021 – 2023*

Experience

NUS & IPAL (CNRS) — *Singapore*

Research Intern, Event-based Vision for Assistive Robots — *Mar 2026 – Jul 2026*

- Cut the trajectory error of DEVO, the state-of-the-art learned event-camera odometry, by 41% on EVIMO2 by designing a 1.0M-parameter causal ConvLSTM that down-weights independently moving objects inside patch selection and differentiable bundle adjustment.
- Showed that end-to-end joint training is key, capturing over 90% of a ground-truth mask's gain vs. 60% when trained separately, and that a label-free variant self-supervised by the bundle adjustment still cuts error by 37%.
- Beat the baseline in 15 of 20 zero-shot evaluations on 6 datasets (EVIMO1/2, MVSEC, EDS, TUM-VIE, VECtor); paper and code release in progress.

Headmind Partners AI — *Paris, France*

AI Engineer Intern — *Jun 2025 – Sep 2025*

- Designed and deployed a multi-agent LLM pipeline where role-played HR, CTO and CEO agents assess each CV, then rank candidates.
- Delivered it as a finished internal tool (OpenAI API, Hugging Face embeddings, Streamlit, unit tests), used to screen 50+ CVs.

Projects

Makeitalive: Animating a Single Landscape Photo — *GitHub*

PyTorch, Diffusers — *Feb 2026 – Mar 2026*

- With a teammate, iterated from an optical-flow U-Net to LoRA adapters (3.3M parameters) to full fine-tuning of Stable Video Diffusion's temporal layers on 2,374 drone clips (12 h on an A100), turning a single photo into a video with cinematic drone-like parallax.

Reinforcement Learning for the Block Blast Puzzle Game — *GitHub*

PyTorch, Gymnasium, Weights & Biases — *Feb 2026 – Mar 2026*

- In a team of 4, increased average survival 4.7× over a greedy baseline (209 vs. 44 moves over 250 games) by replacing DQN and Rainbow with an afterstate value network 63× smaller (42k parameters) using row and column detector convolutions.

edge-mppi: Real-time GPU Model Predictive Control *(in progress)* — *GitHub*

CUDA, C++, Python — *Sep 2026 – Present*

- Modeled a car-like vehicle (kinematic, then dynamic bicycle with tires) and implemented MPPI in NumPy, driving full laps with 1,024 rollouts.
- Now porting it to a fused CUDA kernel, optimized with Nsight, to fit 8,192 rollouts in 20 ms on an RTX 4000 Ada, then benchmark vs. a baseline.

Retrieval Assistant for Emergency Call Handlers — *GitHub*

Python, FAISS, Llama — *Sep 2024 – May 2025*

- In a student team, built an on-premise RAG assistant over a 190-page medical regulation guide, with LLM page routing before FAISS retrieval and source-linked answers, tested in a SAMU call center after interviews and live-call observation with operators.

X-Forum Careers Fair Platform — *xforum.binets.fr*

React, TypeScript, Django, Docker, CI/CD — *Apr 2026 – Present*

- Co-built the portal of École polytechnique's careers fair, serving 250 companies and 3,000 students and processing 160 stand orders: 9-step ordering wizard, passwordless login, DocuSign-signed purchase orders, accounting export, sales-pipeline dashboard.

Skills

Programming	Python, C++, C, CUDA, Java, SQL, OCaml, TypeScript, HTML/CSS
Machine Learning	PyTorch, Hugging Face (Transformers, Diffusers), scikit-learn, XGBoost, pandas
Systems & Tools	Git, Docker, CI/CD, Slurm, Weights & Biases, React, Django, Streamlit
Languages	French (native), English (C1), Spanish (B2)
Leadership	Gendarmerie officer cadet, aide to the commander of a 250-strong company (2023–24) · X-Forum logistics team
Interests	Basketball (university team), BMX racing (10 years, inter-regional), piano and guitar